Supplementary Information

Supplementary Methods

Representative transcript selection from Ribo-seq alignments

For each cellular context, exon annotations were parsed from the GENCODE GTF file to define non-overlapping exon bins from the union of transcript exon boundaries, gene intervals spanning each annotated gene and splice junctions connecting consecutive exons. MANE Select tags were extracted to identify representative isoforms. Ribo-seq reads were counted genome-wide at their 5′ ends, and junction-spanning reads were identified from N operations in the CIGAR string. Exon bins supported by at least two Ribo-seq reads were designated informative features; observed splice junctions provided additional isoform support.

For each gene, a binary feature-by-transcript matrix linked informative features to compatible transcript isoforms. Redundant non-MANE transcripts with identical feature sets and nested non-MANE transcripts whose feature sets were strict subsets of another transcript were removed. An iterative greedy procedure then selected the transcript explaining the largest number of informative features not yet covered. When multiple transcripts shared the same informative features, preference was given to the transcript containing fewer interspersed low-support features. Selection continued until no additional transcript increased informative-feature coverage. MANE Select transcripts were retained irrespective of whether they added a new informative feature.

Nuclease-aware P-site inference

To infer P-sites from Ribo-seq libraries generated under different nuclease conditions, including RNase I and micrococcal nuclease (MNase), we developed a nuclease-aware LightGBM classifier (version 4.6.0) that assigned a read-specific offset from the RPF 5′ end. For each nuclease class, informative fragment lengths and initial offset ranges were identified from read-density peaks around annotated translation initiation or termination sites and from CDS three-nucleotide periodicity. Candidate labels were the offsets between an RPF 5′ end and the corresponding annotated reference site.

For each read, the classifier used one-hot-encoded nucleotide windows spanning 8 nucleotides on either side of both the 5′ and 3′ ends, together with a one-hot representation of fragment length, to predict the candidate offset as a multiclass outcome. Models were fitted with a learning rate of 0.1, up to 2,000 trees, class-balanced weights and early stopping after 100 rounds without improvement. Fragment lengths with strong aggregate periodicity (periodicity score at least 0.55) were assigned a fixed, frame-adjusted offset. For all other informative lengths, the P-site was assigned using the highest-probability offset within the length-specific candidate range. Read counts assigned to the same transcript position were summed to obtain the nucleotide-resolution P-site count profile.

Selection of informative transcript–context profiles

After P-site inference, transcript–context profiles were required to have matched RNA-seq abundance of at least 1.0 reads per million mapped reads (RPM) and sufficient P-site signal. For transcripts with valid annotated CDS coordinates and a CDS longer than 45 nucleotides, P-site depth and coverage were calculated over the CDS. CDS depth was defined as the total number of inferred P-sites divided by CDS length, and CDS coverage as the fraction of consecutive CDS codons containing at least one inferred P-site. CDS P-site depth and coverage were each required to be at least 0.3, and more than 33% of CDS P-sites were required to fall in frame 0.

For transcripts without a valid annotated CDS, P-site depth and coverage were calculated over the full transcript. Full-length depth was defined as the total number of inferred P-sites divided by transcript length, and full-length coverage as the fraction of consecutive three-nucleotide bins containing at least one inferred P-site. Full-length P-site depth and coverage were each required to be at least 0.1; no frame-periodicity criterion was applied.

RNA-abundance normalization and target construction

We transformed each P-site count profile into an RNA-abundance-adjusted relative ribosome occupancy profile. Within each transcript–context profile, non-zero P-site counts were winsorized at the 99.5th percentile. To estimate transcript-level translation relative to RNA abundance, total RPF and RNA-seq counts were each augmented by a pseudocount of 10 and centered-log-ratio (CLR) transformed across retained transcripts within the same cellular context. An ordinary least-squares regression of CLR-transformed RPF counts on CLR-transformed RNA-seq counts was then fitted. The residual, r_t, represented translation not explained by measured RNA abundance, and $\exp(r_t)$ was used as a multiplicative translation-efficiency scale. For transcript t and nucleotide position ℓ, the final target was defined as:

$$y_{tl} = \log\left[1 + \frac{c_{tl}^{(w)}}{\acute{c}_t^{(+)}} \exp(r_t)\right]$$

$$\acute{c}_t^{(+)} = \frac{1}{|P_t^+|} \sum_{l \in P_t^+} c_{tl}^{(w)}, r_t = clr(R_t + 10) - [\beta_0 + \beta_1 clr(A_t + 10)].$$

Here $c^{(w)}_{tl}$ denotes the winsorized P-site count, P_t^+ is the set of positions with a positive winsorized count, $\bar{c}_t^{(+)}$ is the mean over those positions, R_t is the transcript-level RPF count and A_t is the matched RNA-seq count. The regression coefficients β_0 and β_1 were estimated separately within each cellular context. The resulting y_{tl} is dimensionless and represents a relative occupancy signal after adjustment for transcript abundance; it is not an absolute ribosome density, a single-molecule measurement or a site-occupancy probability. This procedure yielded 1.8 million transcript–context profiles from 73 cellular contexts across human, rhesus macaque and mouse. A transcript observed in multiple cellular contexts contributed one profile per context.

Ablation experiments

Structural ablations used the same four-channel nucleotide representation, 384-dimensional token embedding, nucleotide-resolution prediction head, loss functions and chromosome-disjoint data partitions as TRACE. BaseModelLN comprised 12 standard pre-LayerNorm Transformer layers with 16 attention heads, a feed-forward dimension of 768 and dropout of 0.1. It accepted only the RNA sequence and padding mask; expression, cellular-context and species arguments were retained in the software interface for compatibility but were not used in the computation graph. BaseModelConv replaced the Transformer stack with 12 residual one-dimensional convolutional blocks. Each block used pre-LayerNorm, a width-7 convolution from 384 to 384 channels, GELU activation, a 1 × 1 projection back to 384 channels and dropout of 0.1. The convolutional hidden width was selected so that encoder parameter counts differed from BaseModelLN by less than 0.3%. The two sequence-only models shared the same prediction head and optimization procedure.

The cellular-context ablation used the TRACE Transformer backbone under three expression-conditioning strategies. In Zero, the complete expression vector was forced to zero during both training and validation. In Real, the measured 16,840-gene expression vector was supplied without masking, interpolation or added noise. In Expression augment, 10% of training samples received a strictly zero expression vector; among the remaining samples, 30% were multiplied by a scalar sampled uniformly from 0 to 1, continuously interpolating between zero and the measured expression profile. Gaussian noise with a standard deviation of 0.15 was added before scaling. Strictly masked samples remained exactly zero. The Real–Zero comparison quantified the information contributed by the measured cellular environment. TRACE-Zero versus BaseModelLN assessed the effect of the adaptive-normalization and residual-gating parameterization in the absence of measured expression, and BaseModelLN versus BaseModelConv assessed global self-attention relative to a local convolutional encoder.

To assess the effect of cellular-environment diversity, Zero, Real and Expression augment models were trained on matched subsets containing 5, 22 or 40 human cellular environments. The 40-environment set combined 22 tissues and 18 common cell lines. All nine environment-count-by-conditioning combinations were evaluated on the same test set of chromosome-held-out transcripts from 26 uncommon human cell lines that were absent from training. Checkpoint selection and all other training settings followed the TRACE training procedure. Generalization was summarized using transcript-level nucleotide-profile Spearman ρ, periodicity-related performance, CDS-mean signal Spearman ρ and CDS-mean absolute error, with cellular environments weighted equally during aggregation.

Supplementary Table 2 | Capability comparison of TRACE with prior translation models and ORF-identification tools

Model	Full-length RNA sequence	Variable-length input	CDS-annotation-free inference	Raw Ribo-seq-free inference	Single-nt ribosome-density output	Cell-context conditioning	Unified multi-species modeling	Direct TE prediction	Direct ORF identification
TRACE	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Translatomer	No	No	**Yes**	**Yes**	No	**Yes**	No	No	No
RiboMIMO	No	**Yes**	No	**Yes**	No	No	No	No	No
Riboformer	No	No	No	No	No	No	No	No	No
Optimus 5-Prime	No	No	No	**Yes**	No	No	No	**Yes**	No
RiboDecode	No	No	No	**Yes**	No	**Yes**	No	**Yes**	No
RiboNN	**Yes**	**Yes**	No	**Yes**	No	**Yes**	No	**Yes**	No
RiboTIE	**Yes**	**Yes**	**Yes**	No	No	No	No	No	**Yes**
Ribo-TISH	**Yes**	**Yes**	No	No	No	No	No	No	**Yes**
RibORF	**Yes**	**Yes**	No	No	No	No	No	No	**Yes**
TranslationAI	**Yes**	**Yes**	**Yes**	**Yes**	No	No	No	No	**Yes**

Table note. Yes denotes direct support under the operational definitions below; No denotes that the capability is absent or is available only indirectly, through a fixed local window, a separate species-specific model, task-specific retraining, post hoc analysis or a required Ribo-seq input. All columns are phrased as positive capabilities, so Yes under raw Ribo-seq-free inference means that raw or processed Ribo-seq data are not required at inference. A No designation therefore indicates a difference in scope rather than lower overall model quality.

Definitions: Full-length RNA input includes either a complete transcript sequence (5′ UTR, CDS and 3′ UTR) or full-length transcript information reconstructed from a genome FASTA file and GTF/GFF annotation. Variable-length input excludes models restricted to a fixed local sequence window; padding or chunking of biologically variable transcripts is allowed. CDS-annotation-free inference requires no CDS, start-codon, reading-frame annotation or predefined 5′-UTR/CDS boundary at inference; Optimus 5-Prime is therefore classified as No. Raw Ribo-seq-free inference means that neither raw Ribo-seq reads nor processed ribosome-profiling signals, such as mapped RPF counts, P-site profiles or a matched control-condition ribosome-density track, are required as model inputs. Single-nt ribosome-density output refers specifically to a predicted ribosome-density value at each nucleotide, not TIS/TTS probabilities or codon-level profiles. Explicit cell-context conditioning requires a cell-environment representation supplied at inference; fixed cell-type output heads or matched control Ribo-seq tracks do not qualify. Unified multi-species modeling requires one jointly trained framework or checkpoint that models multiple species without species-specific retraining. Direct TE prediction requires TE or a directly corresponding translation-output quantity as a model output. Direct ORF identification requires explicit TIS/ORF discovery or translated-ORF classification.

Feature assignments were checked against the primary publications and official implementations summarized in TRACE_prior_work_comparison.xlsx.